Industrial Internship Report on

Content Management System for a Blog

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was Content Management System for a blog This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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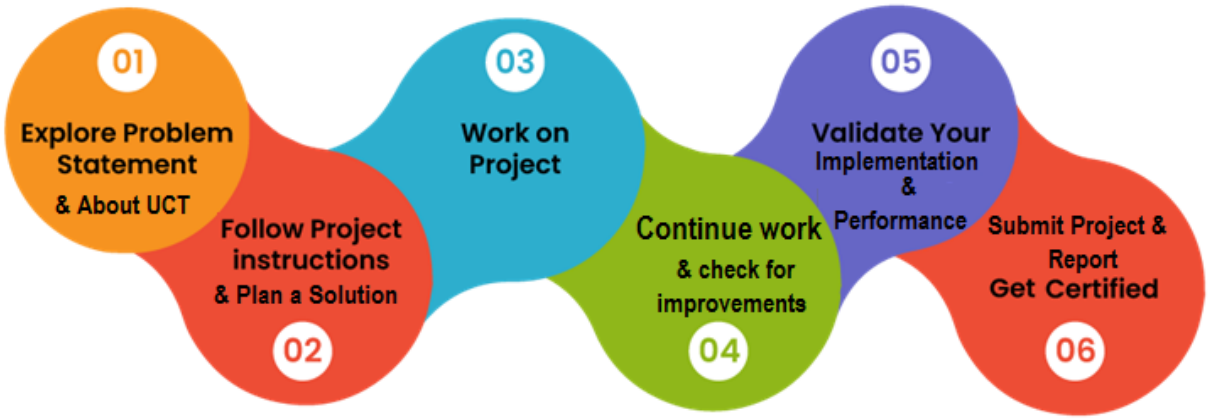
1 Preface

In this 6 weeks of Internship I learn from basic fundamentals to advanced level concepts of Java, JDBC and Full stack development with Quiz. Also create a real-world project Content Management System for a blog.

Internshship is a opportunity after learning to test skills and learn new industry level practical skills. It add an experience of how domain actually works. The objective for internship was to solve real world problems, to have improved job prospects.

UpskillCampus provided an opportunity for learners to build and implement their skills in different categories like python, java, data science and machine learning with Industry partner UniConverge Technologies Pvt. Ltd. for providing study materials, proof offer letter and review system for progress reports.

How Program was planned



My overall experience during this six weeks program was excellent and helpful. I learned technologies like Java, JDBC, Frontend-Backend of Applications. And real-world problems facing in company projects and how to solve them.

Thank to all UpskillCampus and UniConverge Technologies Pvt. Ltd. , who have helped me directly or indirectly.

My message for my juniors and peers is keep learning new skills and keep update yourself. Focus on your interest domains and gain real-world practical knowledge.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

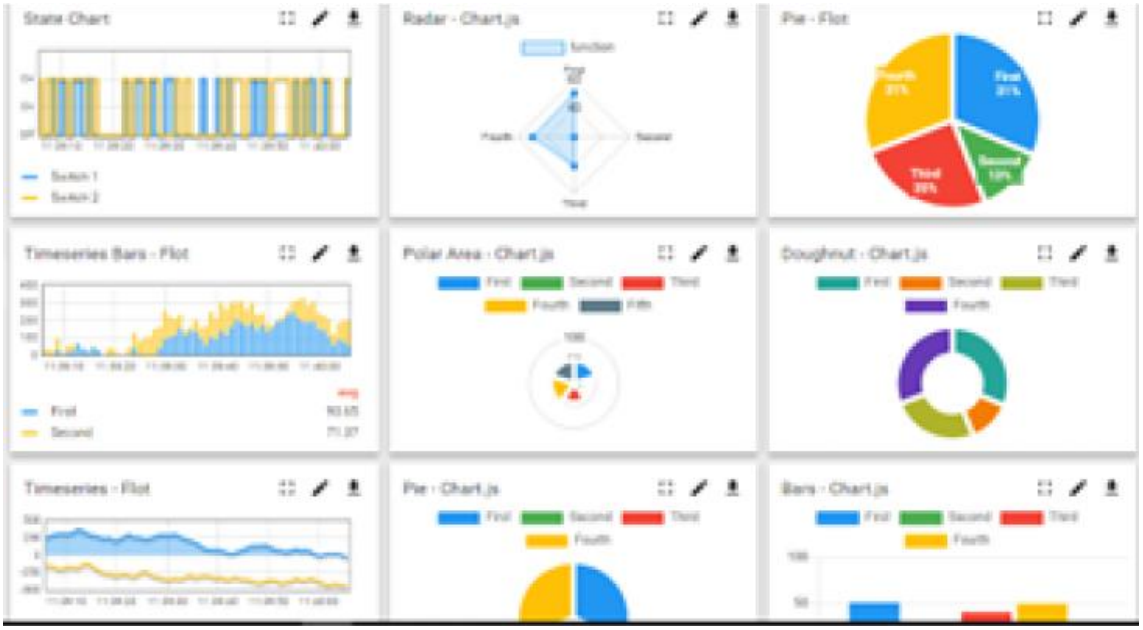
i. UCT IoT Platform (uct Insight)

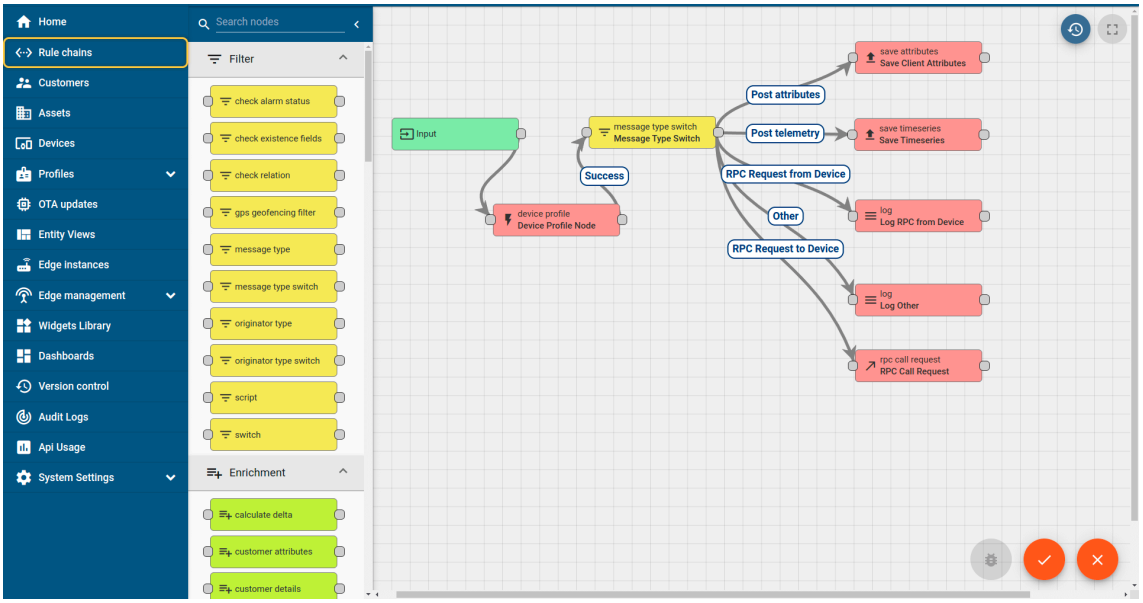
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

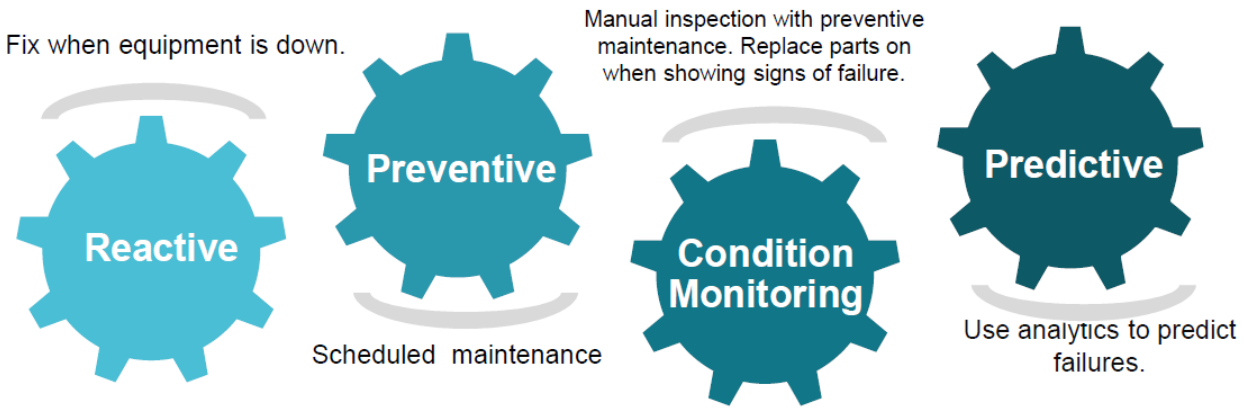
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

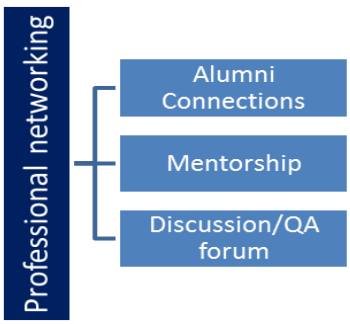
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

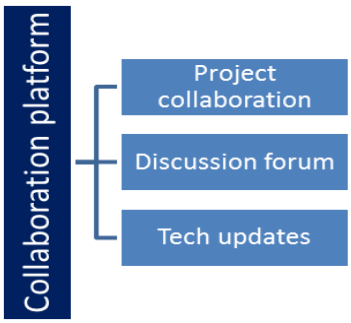


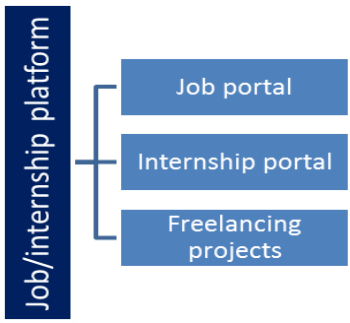
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

In today's digital world, individuals and organizations need an easy way to publish content on the internet without having advanced technical knowledge of web development. However, many existing website development methods require knowledge of programming languages, complex frameworks, and manual coding, which can be difficult for beginners and non-technical users.

The problem addressed in this project is the development of a Content Management System (CMS) for a blog that allows users to easily create, manage, and publish blog posts through a simple and interactive interface. The system enables users to add text and images, organize posts using categories, and interact with readers through a comment system. Additionally, the CMS provides a drag-and-drop image upload feature to simplify content creation.

The goal of this project is to design and implement a user-friendly web application using HTML, CSS, JavaScript, PHP, and MySQL that allows blog owners to manage their website content efficiently without requiring advanced programming skills. The system should support creating posts, uploading images, displaying blog content dynamically from a database, and allowing visitors to read and comment on blog posts.

This project demonstrates how full-stack web technologies can be integrated to build a simple but functional content management system that improves usability and accessibility for content creators.

4 Existing and Proposed solution

Existing Solutions:
There are several content management systems available such as WordPress, Drupal, and Blogger that allow users to create and manage blogs. These platforms provide many features for website creation and content management. However, they can be complex for beginners, require additional setup, and often include many advanced features that may not be necessary for simple blog management. This can make them difficult to understand for new users who only need basic blogging functionality.

Proposed Solution:
The proposed solution is to develop a simple and user-friendly **Blog Content Management System** using HTML, CSS, JavaScript, PHP, and MySQL. The system allows users to create blog posts, upload images, select categories, and manage blog content easily through a clean interface. Blog posts are stored in a database and displayed dynamically on the website.

Value Addition:
The system includes features such as image upload, category selection, comment functionality for readers, and a drag-and-drop image upload option. These features make the system interactive and easy to use while demonstrating full-stack web development concepts.

4.1 Code submission (Github link) :

https://github.com/24amtics009-spec/upskillcampus.git

4.2 Report submission (Github link) :

https://github.com/24amtics009-spec/upskillcampus.git

5 Proposed Design/ Model :

The proposed system follows a simple web application design where users interact with the website interface to create and manage blog content. The process begins when a user opens the blog website and navigates to the post creation page. Here, the user can enter the blog title, select a category, write content, and upload an image using the drag-and-drop or file upload feature.

Once the user submits the form, the data is processed using PHP and stored in a MySQL database. The system then retrieves this stored data and dynamically displays the blog posts on the homepage. Visitors can open individual posts to read the full content and add comments. These comments are also stored in the database and displayed below the respective blog post.

Thus, the system follows three main stages: **content creation**, **data storage**, and **content display**, which together allow efficient management and publishing of blog posts through a user-friendly interface.

5.1 High Level Diagram (if applicable) :

The high-level diagram shows the overall architecture of the Blog Content Management System and how the main components interact with each other.

User / Visitor
|
▼
Web Browser (UI)
(HTML, CSS, JavaScript)
|
▼
Web Server (PHP)
(Process Requests & Logic)
|
▼
MySQL Database
(Store Posts & Comments)
|
▼
Web Server (PHP)
(Retrieve Stored Data)
|
▼
Web Browser (UI)
(Display Blog Content)

Flow Explanation:

User → interacts with the web interface → request is sent to the PHP server → the server processes the request and communicates with the MySQL database → data is retrieved from the database → the blog content is displayed back to the user in the browser.

5.2 Low Level Diagram (if applicable) :

The low-level diagram represents the detailed flow of data and interaction between different modules of the Blog Content Management System.

User
|
▼
Web Browser (HTML, CSS, JavaScript)
|
▼
Post Creation Form
(Title | Category | Content | Image Upload)
|
▼
PHP Backend Processing
(Validation + Image Upload Handling)
|
▼
MySQL Database
(Store Posts, Categories, Comments)
|
▼
PHP Retrieval Scripts
(Get Posts & Comments)
|
▼
Dynamic Web Pages
(Display Blog Posts and Comments)
|
▼
Visitors Read Posts & Add Comments

Flow Explanation:

User → fills blog post form → PHP processes request → data stored in MySQL database → PHP retrieves stored data → blog posts displayed on the website → visitors can read posts and submit comments.

6 Performance Test :

The performance of the Blog Content Management System was evaluated based on factors such as **response time, database performance, and server load**. These constraints are important to ensure that the system works efficiently and provides a smooth user experience.

The system is designed using **HTML, CSS, JavaScript, PHP, and MySQL**, which makes it lightweight and capable of running efficiently on a local server environment like XAMPP. The database structure is kept simple so that blog posts and comments can be stored and retrieved quickly.

During testing, the system was able to create, store, and display blog posts with images and comments without noticeable delays. The response time remained fast for normal usage.

6.1 Test Plan/ Test Cases :

Test Case	Test Description	Input	Expected Result	Actual Result	Status
1	Create new blog post	Title, Category, Content, Image	Blog post should be saved in database and displayed on homepage	Blog post created successfully	Pass
2	Upload blog image	Image file	Image should upload and display with post	Image uploaded successfully	Pass
3	Select category	Technology/Travel/etc.	Selected category should display with post	Category displayed correctly	Pass
4	View blog post	Click "Read More"	Full blog post should open	Post opened correctly	Pass
5	Add comment	Name and comment text	Comment should be saved and displayed under post	Comment displayed correctly	Pass
6	Delete post	Click delete button	Post should be removed from database	Post deleted successfully	Pass

6.2 Test Procedure

First, the system was tested by creating new blog posts by entering the title, selecting a category, writing the content, and uploading an image. The system successfully stored the data in the MySQL database and displayed the post on the homepage.

Next, the blog viewing functionality was tested by clicking on individual posts to check whether the full content and image were displayed properly. The comment feature was also tested by adding user comments under blog posts and verifying that they were saved and displayed correctly.

Finally, the delete functionality was tested by removing blog posts from the system and confirming that they were successfully removed from both the website and the database.

6.3 Performance Outcome :

The performance testing results show that the Blog Content Management System works efficiently under normal usage conditions. The system is able to create, store, and display blog posts along with images and comments without delay.

The response time for loading blog posts and submitting comments was fast when tested on a local server environment using XAMPP. The database operations such as inserting and retrieving blog data were executed successfully with minimal processing time

Overall, the system performed reliably during testing, and the results indicate that the design is capable of handling typical blogging operations effectively.

7 My learnings

Through this project, I learned how to develop a complete web application using **HTML, CSS, JavaScript, PHP, and MySQL**. I gained practical experience in designing user interfaces, handling form inputs, managing databases, and connecting the frontend with the backend.

I also learned how to implement important features such as blog post creation, image upload, category management, and a comment system. Working on this project improved my understanding of full-stack web development and how different technologies work together to build a functional system.

This experience will help in my career growth by strengthening my problem-solving skills and providing practical knowledge of web development, which is useful for developing real-world web applications in the future.

8 Future work scope :

One possible enhancement is adding an **admin login system** to control who can create, edit, or delete blog posts. Another improvement could be implementing an **edit post feature**, allowing users to modify previously published blog content.

Additional features such as **search functionality, category filtering, and like or share options for blog posts** can also improve user interaction. The system can further be enhanced by implementing **dark mode, better UI design, and mobile optimization** to improve the user experience.

future improvements such as **database indexing, caching, and server optimization** can further improve performance and make the system suitable for larger real-world applications.

In the future, the system can also be deployed on a live web server and integrated with advanced technologies such as **REST APIs or cloud databases** to support larger scale applications. These improvements would make the system more powerful and closer to real-world blogging platforms.

- Category filter
- Edit post
- Admin login
- Like button
- Dark mode